Physics - Graduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Among these colors, the one that has the most energy per photon is
   (a) violet
   (b) turquoise
   (c) green
   (d) blue
   (e) red

2. The red glow in the neon tube of an advertising sign is a result of
   (a) fluorescence.
   (b) absorption.
   (c) resonance.
   (d) de-excitation.
   (e) refraction.

3. A heavy rock and a light rock in free fall (zero air resistance) have the same acceleration. The heavy rock doesn't have a greater acceleration because the
   (a) air resistance is always zero in free fall.
   (b) volume of both rocks is the same.
   (c) force due to gravity is the same on each.
   (d) gravitational constant is the same for both rocks.
   (e) force due to gravity is zero in free fall.

4. When small pieces of material are assembled into a larger piece, the combined surface area
   (a) greatly increases
   (b) slightly increases
   (c) is halved
   (d) becomes zero
   (e) greatly decreases

5. A corrective lens of refractive power - 10.00 diopters corrects a myopic eye. What would be the effective power of this lens if it were 12 mm. from the eye?
   (a) -11.00 diopters

(b) -9.50 diopters

(c) -10.50 diopters

(d) -9.00 diopters

(e) -7.93 diopters

True / False (10 marks)

Answer True or False

6. True or False: The minimum kinetic energy of an electron localized within a nuclear radius of 6 x $10^{-}15misapproximately15.9MeV$.

6. True or False: The total collisional frequency for CO 2 at 1 atm and 298 K is approximately 6.28 x $10^{3}4$ m$^{-}3s^{-}1$.

6. True or False: An electron can be effectively accelerated by a magnetic field in a straight line without any additional forces.

7. True or False: When relatively slow-moving molecules condense from the air, the temperature of the remaining air tends to decrease.

8. True or False: The complementary color of blue is cyan, which is often confused with its neighboring color on the spectrum.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the principles of lens magnification by analyzing a scenario where a 40 cm focal length lens projects a 4 cm object onto a screen 20 m away. Calculate the size of the resulting image and explain your reasoning.

12. Discuss the energy transformations that occur during the fission of a uranium nucleus, focusing on the primary forms of energy released and their implications for nuclear physics.

End of Paper